

$$\sin(\pi x) \quad x \quad | -\frac{1}{5} | -\frac{1}{5} | 0 | \frac{1}{5}$$

$$i = 0, 1, 2, 3 \quad n = 3$$

1. Построение интерполяционного полинома в 90-ме. Интерполируемая

$$\begin{aligned} L_3(x) &= f_0 \frac{(x-x_1)(x-x_2)(x-x_3)}{(x_0-x_1)(x_0-x_2)(x_0-x_3)} + f_1 \frac{(x-x_0)(x-x_2)(x-x_3)}{(x_1-x_0)(x_1-x_2)(x_1-x_3)} + f_2 \frac{(x-x_0)(x-x_1)(x-x_3)}{(x_2-x_0)(x_2-x_1)(x_2-x_3)} + f_3 \frac{(x-x_0)(x-x_1)(x-x_2)}{(x_3-x_0)(x_3-x_1)(x_3-x_2)} \\ &= \sin\left(-\frac{\pi}{4}\right) \frac{(x+\frac{1}{5})(x-0)(x-\frac{1}{5})}{(-\frac{1}{5}+\frac{1}{5})(-\frac{1}{5}-0)(-\frac{1}{5}-\frac{1}{5})} + \sin\left(-\frac{\pi}{5}\right) \frac{(x+\frac{1}{5})(x+\frac{1}{5})(x-\frac{1}{5})}{(-\frac{1}{5}+\frac{1}{5})(-\frac{1}{5}-0)(-\frac{1}{5}-\frac{1}{5})} + \sin(0) \frac{(x+\frac{1}{5})(x+\frac{1}{5})(x-\frac{1}{5})}{(0+\frac{1}{5})(0+\frac{1}{5})(0-\frac{1}{5})} + \sin\left(\frac{\pi}{4}\right) \frac{(x+\frac{1}{5})(x+\frac{1}{5})(x-0)}{(\frac{1}{5}+\frac{1}{5})(\frac{1}{5}+\frac{1}{5})(\frac{1}{5}-0)} = \\ &= \frac{48\sqrt{2}x^3 - 40\sqrt{2}x^2 - 2\sqrt{2}x + 27x - 432}{5} + \frac{48\sqrt{2}x^3 + 20\sqrt{2}x^2 + 2\sqrt{2}x - 288\sqrt{2}x^3 - 8\sqrt{2}x - 432}{5} = \\ &= \frac{x^3(288\sqrt{2} - 432) + x(27 - 8\sqrt{2})}{5} \end{aligned}$$

2. Проверка

$$a) x = -\frac{1}{5} \Rightarrow f(x) = \frac{-\frac{1}{5}(288\sqrt{2} - 432) - \frac{1}{5}(27 - 8\sqrt{2})}{5} = -\frac{\sqrt{2}}{2} = \sin\left(-\frac{\pi}{4}\right)$$

$$б) x = -\frac{1}{5} \Rightarrow f(x) = \frac{-\frac{1}{5}(288\sqrt{2} - 432) - \frac{1}{5}(27 - 8\sqrt{2})}{5} = -\frac{1}{2} = \sin\left(-\frac{\pi}{5}\right)$$

$$в) x = 0 \Rightarrow f(x) = 0 = \sin(0)$$

$$г) x = \frac{1}{5} \Rightarrow f(x) = \frac{\frac{1}{5}(288\sqrt{2} - 432) + \frac{1}{5}(27 - 8\sqrt{2})}{5} = \frac{\sqrt{2}}{2} = \sin\left(\frac{\pi}{4}\right)$$

3. Оценка остаточного члена для точек $-\frac{1}{5}$ и $\frac{1}{5}$

$$a) x^0 = -\frac{1}{5}$$

$$\omega\left(-\frac{1}{5}\right) = \left|-\frac{1}{5} + \frac{1}{5}\right| \left|-\frac{1}{5} + \frac{1}{5}\right| \left|\frac{1}{5}\right| \left|\frac{1}{5} + \frac{1}{5}\right| = \frac{-3}{20000}$$

$$f(x) = \sin(\pi x), x \in \left[-\frac{1}{5}, \frac{1}{5}\right]$$

$$M_4 = \max |f^{(4)}(x)| \quad \begin{aligned} f^{(1)}(x) &= \pi \cos(\pi x) & f^{(3)}(x) &= -\pi^3 \cos(\pi x) \\ f^{(2)}(x) &= -\pi^2 \sin(\pi x) & f^{(4)}(x) &= \pi^4 \sin(\pi x) \end{aligned} \Rightarrow M_4 = \frac{\pi^4 \sqrt{2}}{2}$$

$$\frac{\pi^4 \sqrt{2}}{2} \frac{1}{4!} \frac{3}{20000} \approx 0,00086$$

$$|f(x^0) - L_3(x^0)| = \left| \sin\left(-\frac{\pi}{5}\right) + \frac{88\sqrt{2} + 243}{625} \right| \approx 0,786$$

$$б) x^0 = \frac{1}{5}$$

$$\omega\left(\frac{1}{5}\right) = \left|\frac{1}{5} + \frac{1}{5}\right| \left|\frac{1}{5} + \frac{1}{5}\right| \left|\frac{1}{5}\right| \left|\frac{1}{5} - \frac{1}{5}\right| = \frac{-33}{20000}$$

$$M_4 = \frac{\pi^4 \sqrt{2}}{2} \Rightarrow \frac{\pi^4 \sqrt{2}}{2} \frac{1}{4!} \frac{33}{20000} \approx 0,0047$$

$$|f(x^0) - L_3(x^0)| = \left| \sin\left(\frac{\pi}{5}\right) - \frac{88\sqrt{2} + 243}{625} \right| \approx 0,00014$$

4. График погрешности

$$r(x) = f(x) - L_3(x) = \sin(\pi x) - \frac{x^3(288\sqrt{2} - 432) + x(27 - 8\sqrt{2})}{5}$$

